

Neural networks and convolutions

Neural networks compose simple parameterized operations; convolutions reuse local detectors across an image.

15–20 MINUTES

Prerequisite: learning loop

PATTERN-TRACING EXERCISE

1 • Unit

Weighted sum plus activation.

2 • Layer

Many features in parallel.

3 • Convolution

Shared local weights.

4 • Hierarchy

Features compose across layers.

From one score to a network

A basic neural-network unit computes a weighted sum, adds a bias, and passes the result through an activation function. A layer computes many such outputs. Stacking layers lets later computations combine patterns detected earlier.

$$\text{activation} = f(\text{weights} \cdot \text{inputs} + \text{bias})$$

The activation function makes the composition nonlinear. Without nonlinearities, stacking linear layers would still be equivalent to one linear transformation and would not express the same variety of decision boundaries.

Central mental model

Early layers respond to local patterns; later layers combine them into representations useful for the task. Training adjusts all the reusable detectors together.

Why ordinary dense layers are awkward for images

A dense layer gives every input value a separate connection to every output. That ignores two useful facts about images: nearby pixels are related, and the same pattern can appear anywhere. A **convolution** applies one small kernel at every valid location. The weights are shared across space.

Weight sharing reduces the parameter count and lets one learned edge detector respond on the left or right of an image. The output for one kernel is a feature map. Multiple kernels produce multiple output channels.

Stride, padding, and receptive field

Stride controls how far the kernel moves between outputs. A larger stride reduces spatial resolution. **Padding** adds border values so edge locations can be processed and size can be controlled. The **receptive field** of an activation is the input region that can influence it. It grows through stacked layers.

Lower resolution can make computation cheaper and expose larger context, but precise localization becomes harder. Detection networks therefore combine information across resolutions.

Browser exercise: trace a local detector

Imagine a 3×3 kernel trained to respond to a vertical brightness transition. You slide the same kernel across a 100×100 image.

What does weight sharing buy us?

What does one output cell describe?

Applied to our cubes

GENERAL PRINCIPLE A detector backbone builds spatial feature maps; later detection components use them to predict object categories and locations at multiple scales.

ARTIFACT-SPECIFIC FACT The current exported model yields predictions associated with three spatial resolutions. The exact architecture and 8400 candidate count belong to this export, not to every convolutional detector.

Project anchor: model contract.

Check your understanding

What is the key spatial idea behind convolution?

- ☐ Separate weights for every pixel
- ☐ Shared weights across image locations
- ☐ One label for each image

Explain it back

Explain why convolution is a good fit for images. Include locality, weight sharing, feature maps, and how deeper layers obtain larger context.

Primary sources

- D2L: From fully connected layers to convolutions.
- Deep Learning, Chapter 9 — convolutional networks.